

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-007 - Lakwa-007

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000076
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2001-10-03 / 2001-11-23
Measured / TVD	2477 m / 2447 m
Trajectory	Directional; inc 26.1 deg; az 152.2 deg
Formation Interval	1378-1972 m
Lithology	Fluvial sandstone with clay streaks
Pore Pressure Reference	2529 psi
Fracture Gradient	14.53 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (55.1/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	619	Surface
9 5/8 in casing	1436	Intermediate
7 in liner/casing	2179	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2001-11-05 1667 m	Mud Loss	Critical	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2001-11-06 1705 m	Kick	Critical	Pit gain and flow with pumps off were observed. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2001-11-18 2254 m	Mud Loss	Critical	Active pit volume declined during stable pump rate. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2001-10-11 502 m	NPT	High	ROP dropped to zero during equipment unavailability. Mitigation: Use preventive maintenance triggers from repeated downtime patterns. Lesson: Consistent NPT coding enables meaningful cross-well comparison.
2001-10-29 1376 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.
2001-11-12 1991 m	Kick	High	Pit gain and flow with pumps off were observed. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Maintain mud-weight window using offset pressure evidence.
2001-10-22 1060 m	Torque Spike	Medium	High torque developed while WOB remained elevated. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.

Completion Summary

The synthetic well reached 2477 m MD in the Tipam interval. The NWIS knowledge record links 9 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.